

Member ID: _____

Time: _____

Rank: _____



C# PROGRAMMING

(330)

STATE 2022

Production:

Program 1: World Capital Quiz Creator _____ (540 points)

TOTAL POINTS _____ (540 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

World Capital Quiz Creator

In this exercise you will create a C# Windows Form Application that generates a multiple-choice quiz by selecting a certain number of questions and answers from a text file. While the number of questions selected will stay the same for each game, the questions should be randomly selected from the list in the text file every time the game is run.

When started, the program will display the first question in its randomly selected list. Once the user has clicked the radio button corresponding to their selected answer and clicked the “Submit” button, the answers are highlighted (green = the single correct answer, red = all other incorrect answers) and the user can click the “Next” button to advance to the next question. After the user has submitted their answer for the last question and clicked “Next”, a dialog box appears telling the user their score and asking them whether or not they wish to play again.

World Capital Quiz

What is the capital of Germany?

☒ Berlin

☐ Munich

☐ Brandenburg

☐ Cologne

Submit

Next

Requirements:

1. You must create a C# Windows Form Application named CS _330_ContestantNumber, where ContestantNumber is your BPA assigned contestant number (including dashes). For example, CS_123_45_6789_0000.
2. Your contestant number must appear as a comment at the top of the main source code file.
3. The main form must not have minimize or maximize buttons in the Form Handle.
4. The main form must have text “World Capital Quiz” for its title. The form must start in the center of the screen and must have a Fixed Dialog setting so the size cannot be adjusted.
5. The main form must contain the following items:

UI Object	Name	Text	Font Size	Size, Width	Size, Height	Multi line	Read only
Textbox	txtQuestion		29pt	590px	97px	true	true
Radio Button	rdoAnswer1		14pt				
Radio Button	rdoAnswer2		14pt				
Radio Button	rdoAnswer3		14pt				
Radio Button	rdoAnswer4		14pt				
Button	btnSubmit	Submit	20pt	166px	40px		
Button	btnNext	Next	14pt	113px	40px		

6. When the program first loads, the main window needs the following (see pictures for reference):
 - a. txtQuestion should be place in the top left corner of the window.
 - b. The answer radio buttons should be also be left-aligned
 - c. btnSubmit should be center-aligned at the bottom of the window
 - d. btnNext should be right-aligned and also at the bottom of the window.
7. When the program first loads, the quiz questions and answers should be read from a text file named “quizquestion.txt”
 - a. Each line of the text file should represent a different question and its answers. Every question must have exactly four possible answers and only one answer can be considered correct. For best results it is recommended to just use the example question provided in the picture of the example text file. Each line should be formatted as follows (see pictures for example text file):

<QuestionText>,<Answer1>,<Answer2>,<Answer3>,<Answer4>,<CorrectAnswerNumber>

- i. If a line in the text file is not formatted correctly, display a MessageBox with an error message and exit the application (see pictures section for reference)

- ii. If the correct answer number in a line is not a number, display a MessageBox with an error message and exit the application (see pictures section for reference)
- iii. If the correct answer number in a line is not between 1-4, display a MessageBox with an error message and exit the application (see pictures section for reference)
- b. Each line of the text file should be read and parsed as an object of a class called “Question”. The class must contain all the information related to a question, including the question text, the possible answers, and which answer is correct. The class must also contain a constructor which takes a line from quizquestion.txt, parses out all question/answer information, and stores it in the appropriate properties.

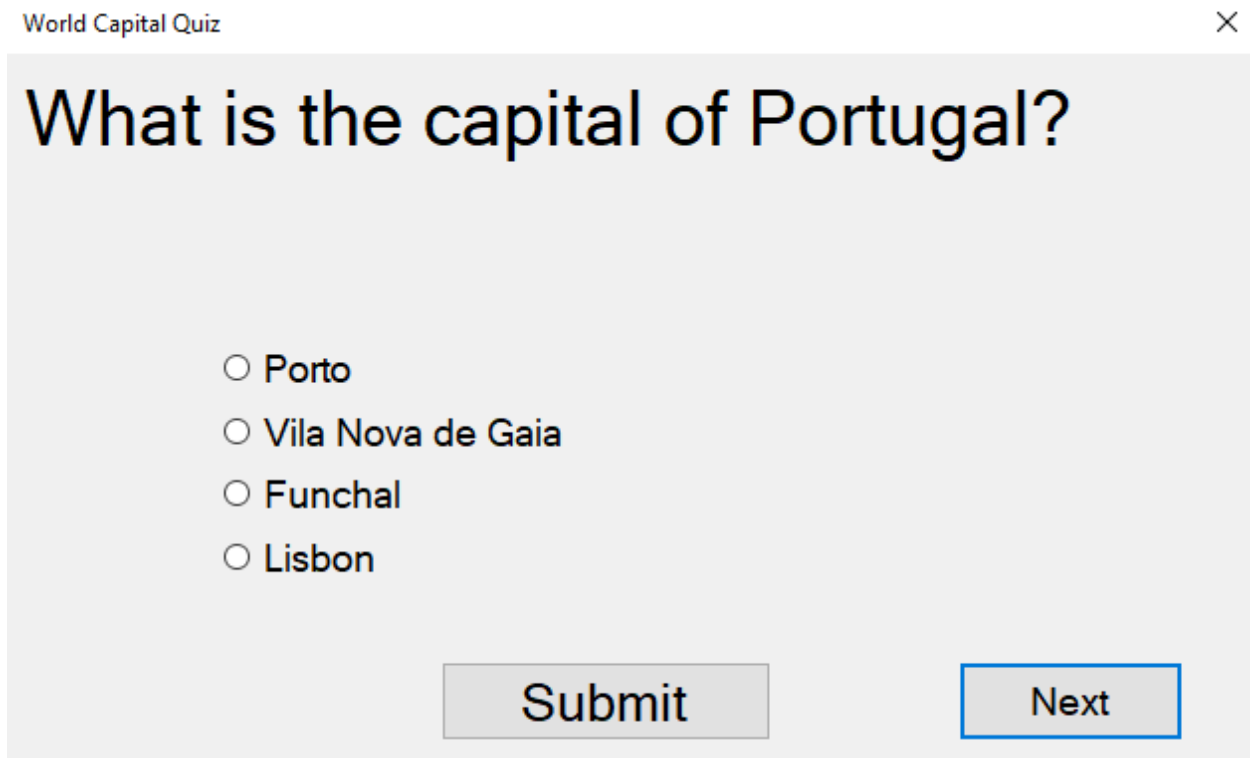
Question
- QuestionText: string
- Answers: IList<string>
- CorrectAnswerIndex: int
- Question(string inputFile)

- c. From the list of newly created Question objects, randomly select three question to use in the quiz. Finally, load the first question by replacing the question text and answer radio button text with the data for the first question.
8. The “Submit” button must use a click event called “SubmitButton_Click” which performs the following functions:
- a. Retrieves the user-selected answer text as specified by the user-selected radio button
 - i. If none of the answer radio buttons are selected, nothing should happen when the user clicks the Submit button
 - b. Change the BackColor of the answer radio button to reveal the correct and incorrect answers. For the correct answer, use Color.DarkGreen. For the incorrect answers, use Color.DarkRed
9. The “Next button must use a click event called “NextButton_Click” which performs the following functions:
- a. If the user has answered the current question and there are more questions left in the quiz, load the next question. This can be done by replacing the question text and answer radio button text with the data for the next question. The BackColors for each radio button should also be reset to Color.Transparent.
 - i. If the user hasn’t submitted their answer to the current question, do nothing
 - b. If the user has submitted their answer and the current question is the last one in the quiz, end the quiz. This can be done by showing two dialog boxes, the first

telling the user their score and the second asking the user if they want to play again (see pictures for reference).

- i. If the user selects “Yes”, pull three random questions from the list and restart the game.
- ii. If the user selects “No”, exit the application.

Sample Pictures:



World Capital Quiz

What is the capital of Portugal?

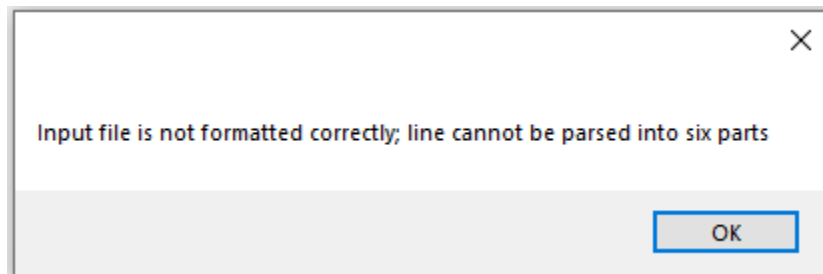
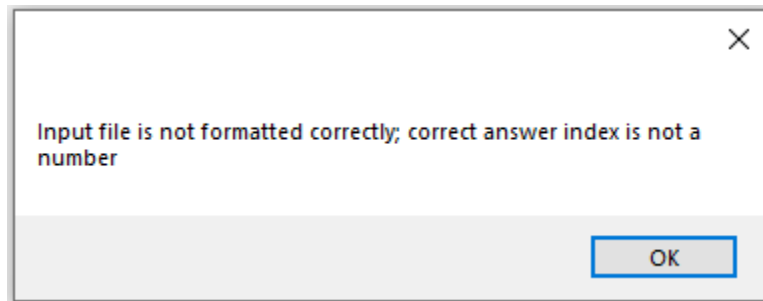
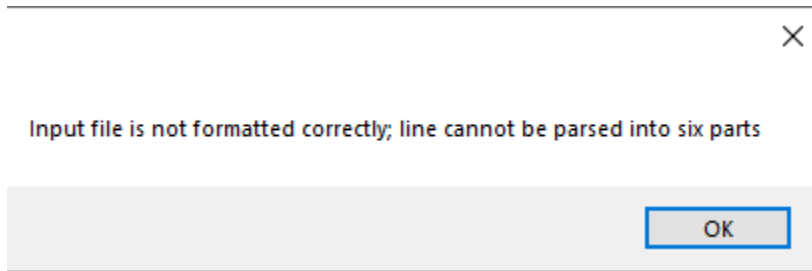
- ☐ Porto
- ☐ Vila Nova de Gaia
- ☐ Funchal
- ☐ Lisbon

Submit Next

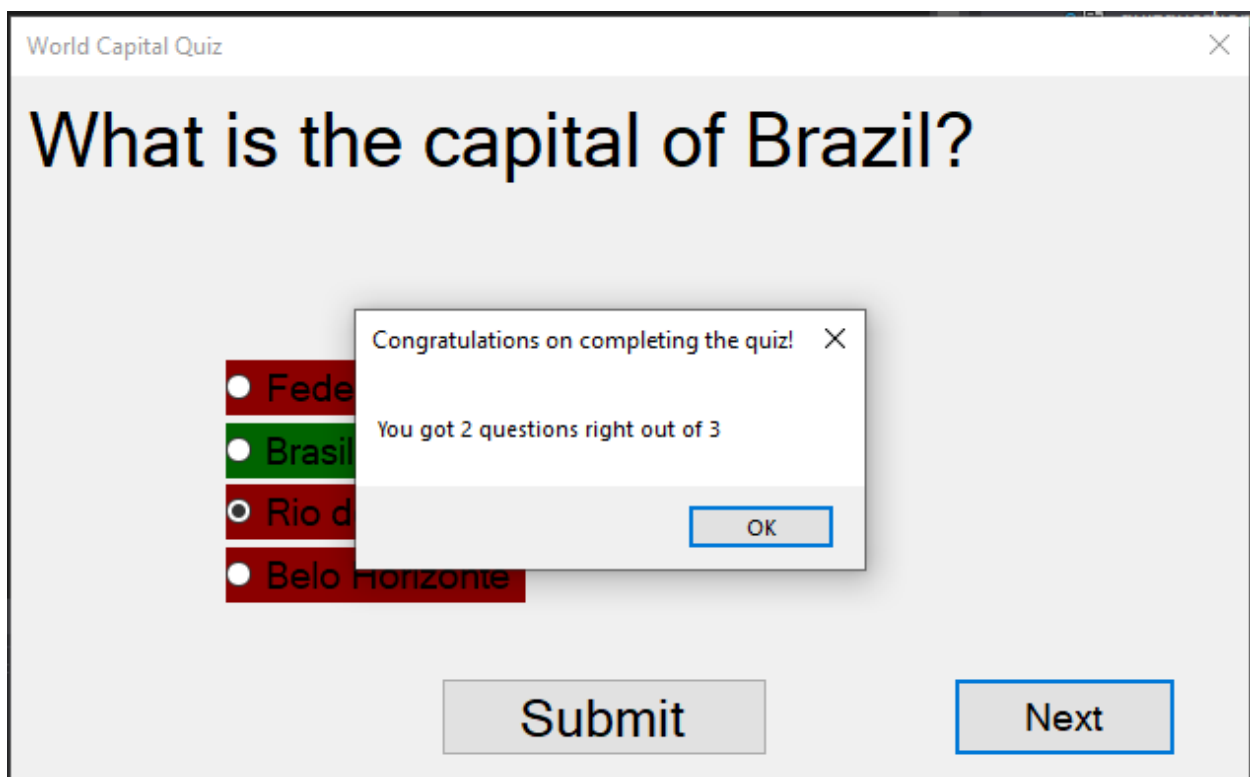
1: Example of the initial layout of the window

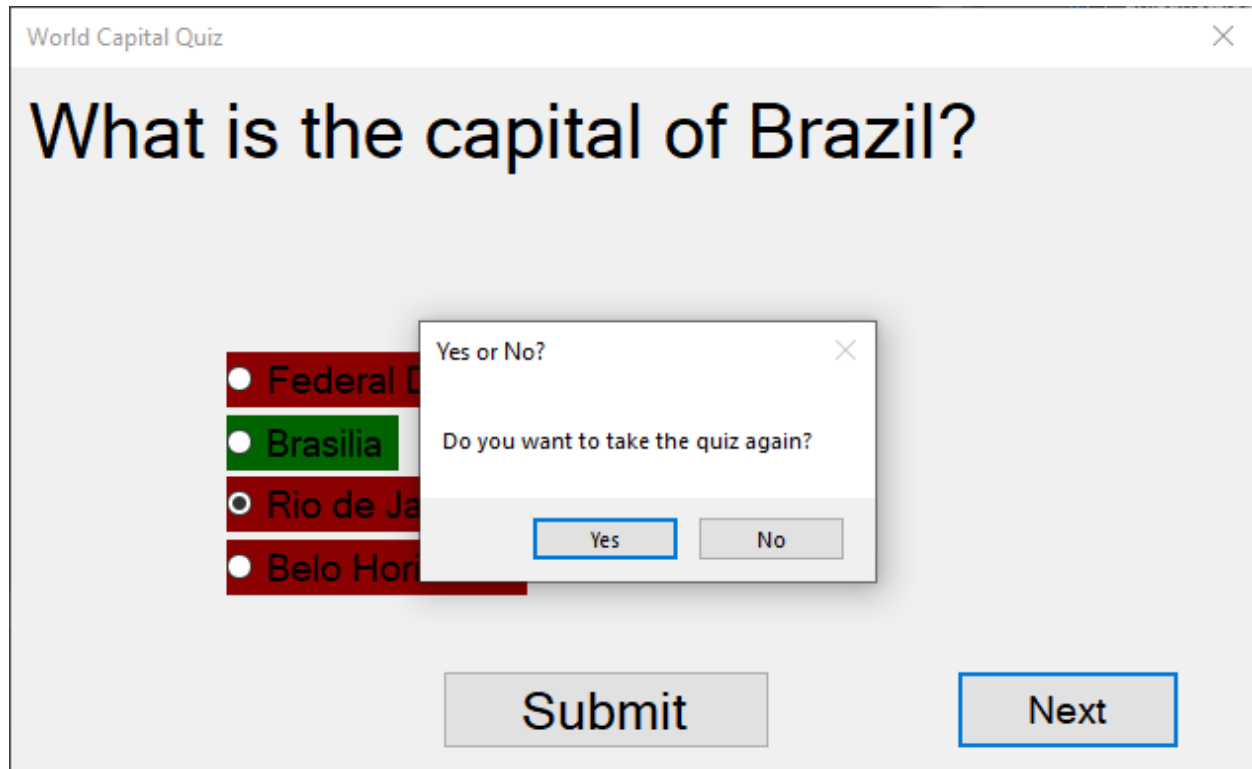
```
What is the capital of Germany?,Berlin,Munich,Brandenburg,Cologne,1
What is the capital of China?,Hangzhou,Yichang,Beijing,Nanning,3
What is the capital of Portugal?,Porto,Vila Nova de Gaia,Funchal,Lisbon,4
What is the capital of New Zealand?,Wellington,Auckland,Christchurch,Melbourne,1
What is the capital of Brazil?,Federal District,Brasilia,Rio de Janeiro,Belo Horizonte,2
What is the capital of Russia?,Saint Petersburg,Moscow,Novosibirsk,Kazan,2
```

2: Example of the quizquestions.txt file



3: Examples of the error message displayed for invalid text file errors





3: Examples of the message boxes displayed when the quiz ends

You will have ninety (90) minutes to complete your work.

Your name and/or school name should not appear on any work you submit for grading.
Remember to use your contestant # where appropriate.

Submit a copy your entire solution/project to the flash drive provided. You must submit your entire solution/project so that the graders may open your project to review the source code and/or build and execute your solution/project. **Submissions that do not contain source code will not be graded.**

Development Standards

- Standard name prefixes must be utilized for variables.
- All subroutines, functions, and methods must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).

Your application will be scored on the following criteria:

Solution and Project

The project is present on the flash drive _____ 10 pts
The project is named according to the naming conventions _____ 10 pts

Program Execution

Code copied to USB drive and program runs from USB _____ 20 pts

If the program does not execute, then the remaining items in the program executive section receive a score of zero.

The title of the form is “World Capital Quiz “ _____ 10 pts
The main form does not have minimize or maximize buttons
in the Form Handle _____ 10 pts
The form’s start position is the center of the screen and it cannot be resized _____ 10 pts
The window has the correct buttons, text boxes, radio buttons and
labels based upon the Figures _____ 30 pts
The program reads all lines from the quizquestions.txt file _____ 50 pts
The program parses each line as a Question object _____ 50 pts
The program **randomly** selects three questions when starting a quiz _____ 50 pts
The program correctly loads the first question upon program start _____ 30 pts
Clicking “Submit” correctly highlights answer text if user answered _____ 20 pts
Clicking “Submit” does nothing if user has not answered _____ 10 pts
Clicking “Next” correctly loads the next question _____ 20 pts
Clicking “Next” resets answer background color to Color.Transparent _____ 10 pts
Clicking “Next” does nothing if user hasn’t submitted answer _____ 10 pts
Clicking “Next” on last question displays ending Message Boxes _____ 20 pts
First ending Message Box gives user their quiz score _____ 20 pts
Second ending Message Box restarts quiz if “Yes” is selected _____ 10 pts

Second ending Message Box ends quiz if “No” is selected _____ 10 pts
The program selects three new random questions upon restart _____ 30 pts

Source Code Review

Code is commented at the top, for each function, and as needed _____ 10 pts
Code uses reasonable and consistent variable naming conventions _____ 10 pts
Code for “SubmitButton_Click” method is present _____ 30 pts
Code for “NextButton_Click” method is present _____ 30 pts
Code for reading questions and answers from text file is present _____ 10 pts
Code for error handling text file parsing errors is present _____ 20 pts
Code for parsing each text file line as a Question object is present _____ 10 pts
Code for defining Question class is present _____ 10 pts
Code for randomly selecting three questions for quiz is present _____ 10 pts
Code for ending the quiz correctly is present _____ 10 pts

Total Points: _____ / 540 pts